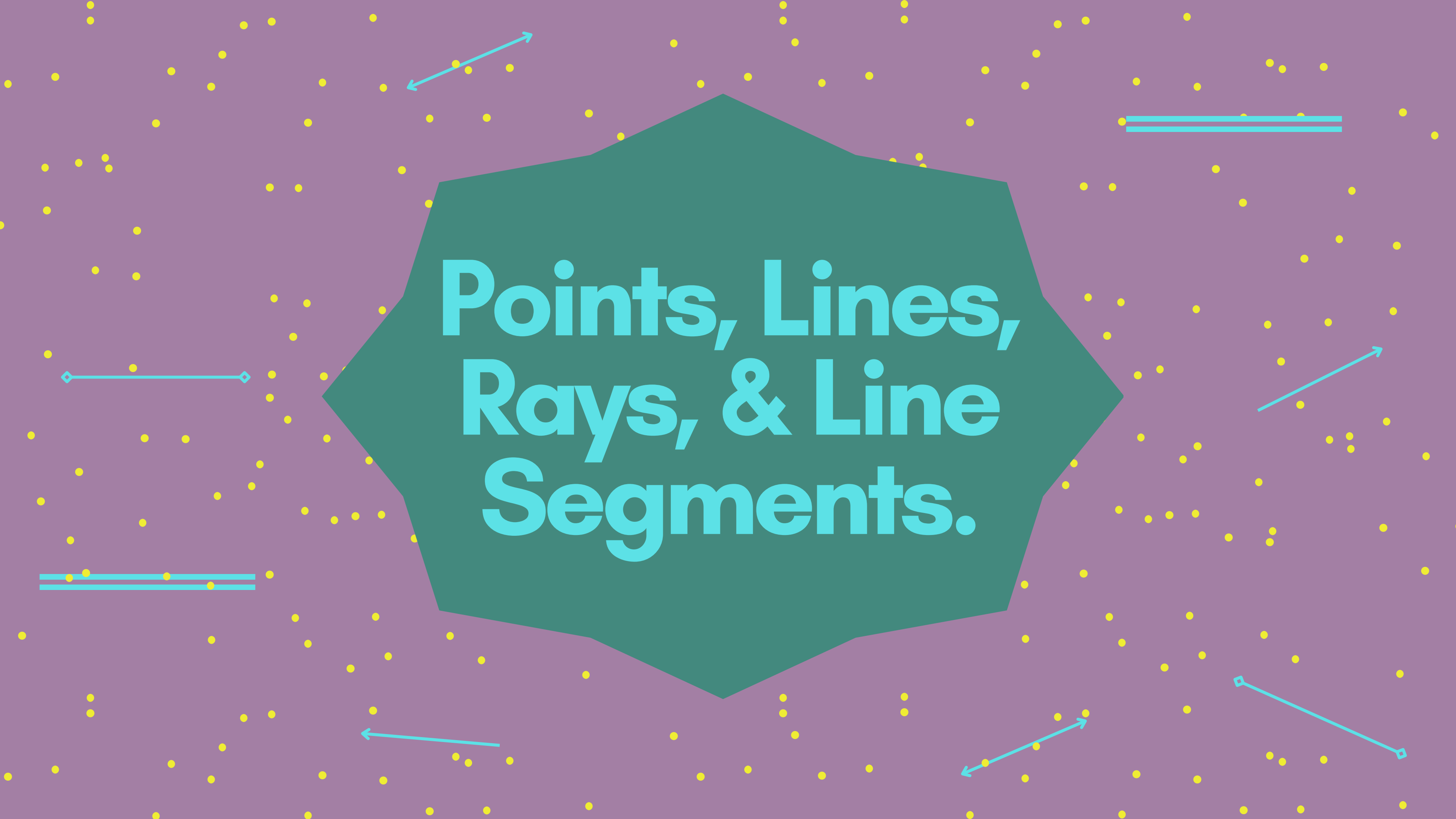


The background is a purple field filled with small yellow dots. Scattered throughout are several geometric elements: a horizontal line with arrows at both ends, a horizontal line with small diamond markers at both ends, a horizontal line with double parallel lines above and below it, a diagonal line with an arrow at one end, a diagonal line with a small square marker at one end, and a diagonal line with arrows at both ends.

# Points, Lines, Rays, & Line Segments.

# Point

Points are the most basic elements in geometry. They are represented by a dot and have no size or dimension. A point is simply a location in space and is denoted by a capital letter, such as point A or point B.

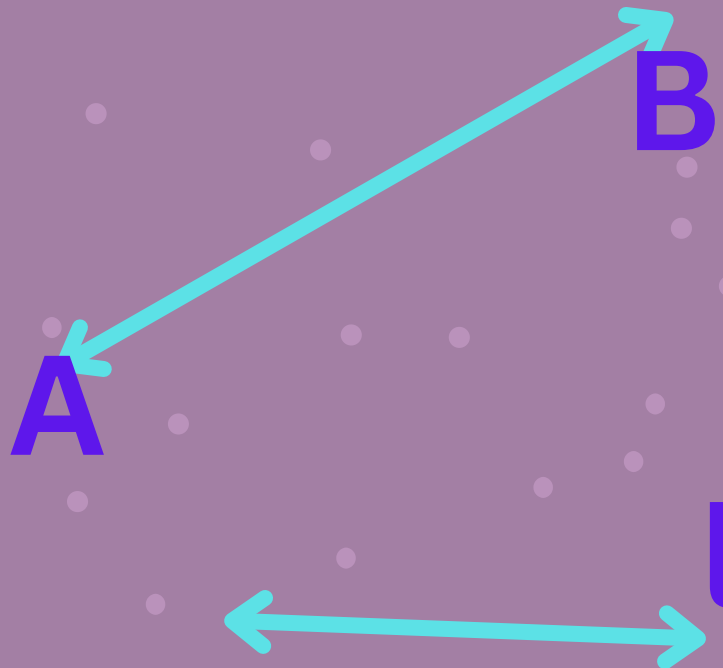


B

A

# Line

A line is a straight path that extends indefinitely in both directions. It is made up of an infinite number of points and has no thickness. In geometry, a line is denoted by naming two points on the line and drawing a line segment between them, or by a single lowercase letter, such as line AB or line l.



# Ray



A ray is a part of a line that has one endpoint (called the endpoint of the ray) and extends infinitely in one direction and has no defined length or width. It is essentially a half-line. Rays are named by their endpoint and another point on the ray that is not the endpoint, such as ray AB.

# Line Segment



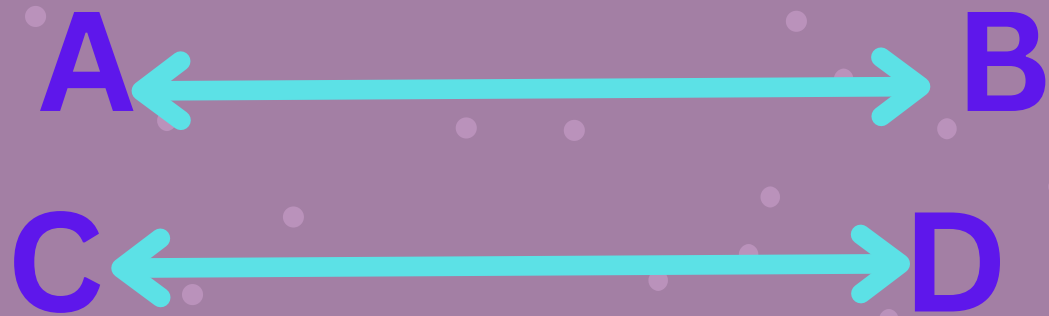
A line segment is a portion of a line that is bounded by two distinct endpoints. Unlike a line, a line segment has finite length. It is named by its two endpoints, such as segment AB or segment CD.



# In summary :

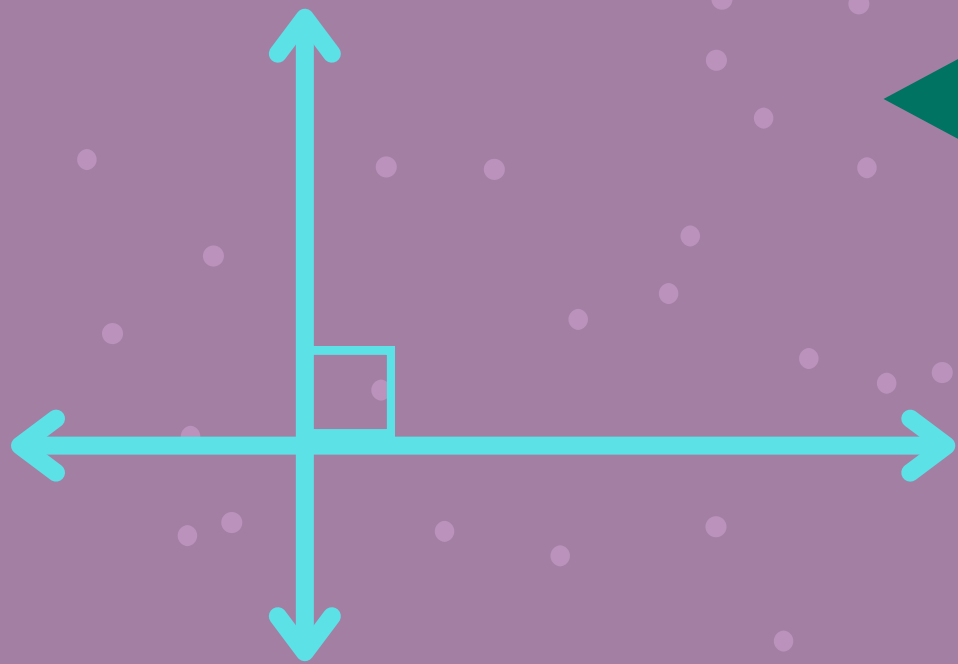
- Points are individual locations in space. 
- Lines are straight paths that extend indefinitely in both directions. 
- Line segments are finite portions of a line with two distinct endpoints. 
- Rays are parts of a line that have one endpoint and extend infinitely in one direction. 

# Parallel Lines



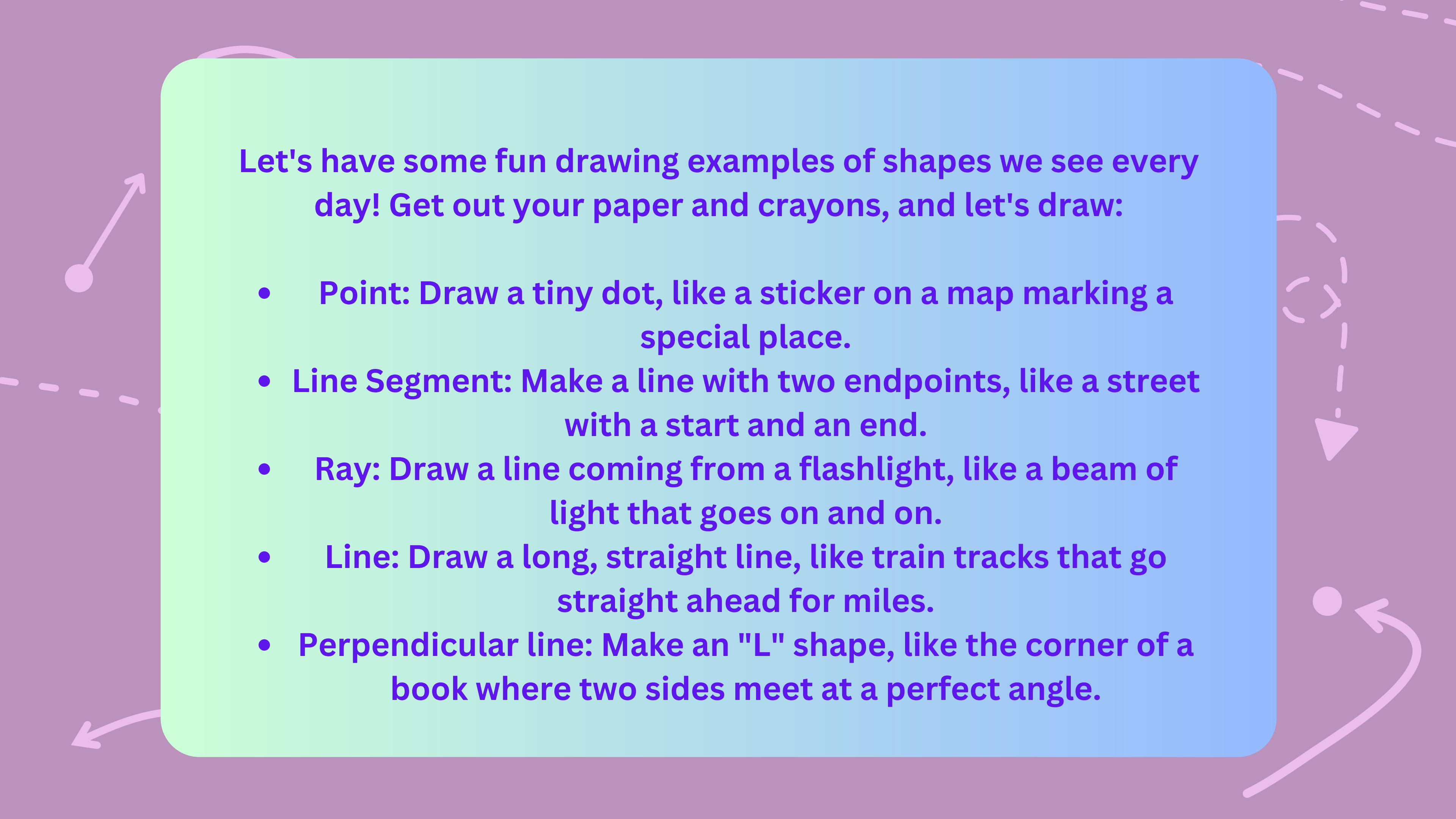
Parallel lines are two or more lines that never meet, no matter how far they are extended. They remain the same distance apart from each other at all points. In other words, they have the same slope and will never intersect. Think of railroad tracks, which are a classic example of parallel lines.

# Perpendicular Lines:



Perpendicular lines are two lines that intersect at a right angle (90 degrees). This means that if you were to extend them infinitely in both directions, they would form a perfect "L" shape where they meet. Picture the corners of a square or rectangle, where the sides meet at right angles. This is a common example of perpendicular lines.



The background is a solid purple color. It features several decorative elements: a dashed white line on the left, a solid white arrow pointing up and to the right, a dashed white arrow pointing down and to the right, a solid white arrow pointing down and to the left, and a solid white arrow pointing up and to the left. There are also some dashed white lines forming a path on the right side.

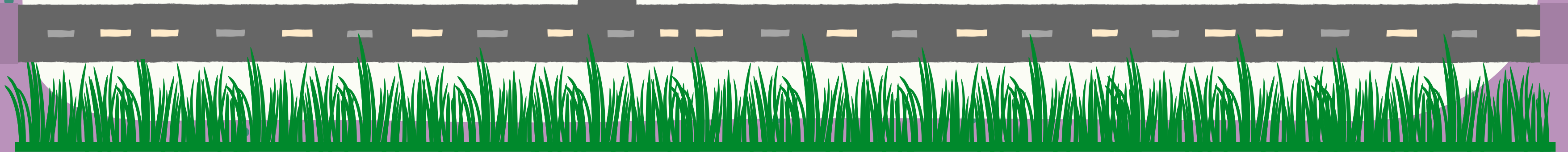
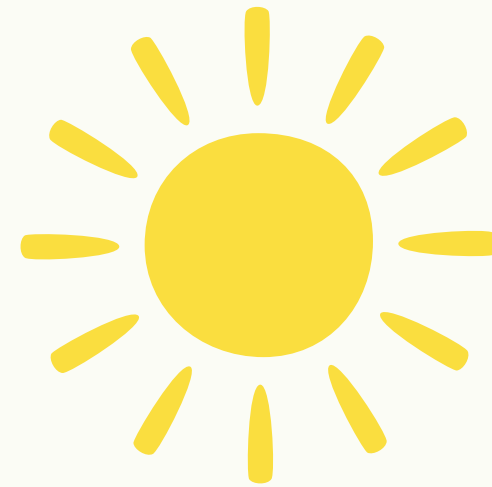
**Let's have some fun drawing examples of shapes we see every day! Get out your paper and crayons, and let's draw:**

- **Point:** Draw a tiny dot, like a sticker on a map marking a special place.
- **Line Segment:** Make a line with two endpoints, like a street with a start and an end.
- **Ray:** Draw a line coming from a flashlight, like a beam of light that goes on and on.
- **Line:** Draw a long, straight line, like train tracks that go straight ahead for miles.
- **Perpendicular line:** Make an "L" shape, like the corner of a book where two sides meet at a perfect angle.

# Activity time

Find the following in the picture:

- a point
- a line segment
- a ray
- a line
- parallel lines
- perpendicular lines



# Key Answers

